VANISHING OF HIGHER DIRECT IMAGES FORCES THE PUSHFORWARD OF A RELATIVELY PERFECT MODULE TO BE LOCALLY FREE WITH BASE-CHANGE COMPATIBILITY

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Editorial note. Input lemma-base-change-tensor-perfect is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Perfectness and base change for the derived pushforward. lemma-base-change-tensor-perfect (same chapter, NOT carded, declared external). E = Rf_*F is perfect and its formation commutes with arbitrary base change.
- Perfect modules of finite presentation are finite locally free. More on Algebra, lemma perfect; Algebra, lemma finite-projective (cross-chapter, named). H^0(E) is finite locally free once flat and perfect.
- Flat cokernel criterion. Algebra, lemma cokernel-flat (cross-chapter, named). Checking O_S -> H^0(E) after tensoring with residue fields suffices.

Source reference. Original reference 00NX: focused-seeds/geometry/S9-perf2-vanishing-implies-locally-free/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0A1H: sources/stacks-perfect/pinned-original-source.tex, source lines 7614–7614. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0658: focused-seeds/geometry/S9-perf2-vanishing-implies-locally-free/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of finite presentation. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation, flat over S with support proper over S. If $R^i f_* \mathcal{F} = 0$ for $i > 0$, then $f_* \mathcal{F}$ is locally free and its formation commutes with arbitrary base change (see proof for explanation).*

Proof. By Lemma 0A1H the object $E = Rf_* \mathcal{F}$ of $D(\mathcal{O}_S)$ is perfect and its formation commutes with arbitrary base change, in the sense that $Rf'_*(g')^* \mathcal{F} = Lg^* E$ for any

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cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

of schemes. Since there is never any cohomology in degrees < 0, we see that E (locally) has tor-amplitude in $[0, b]$ for some b. If $H^i(E) = R^i f_* \mathcal{F} = 0$ for $i > 0$, then E has tor amplitude in $[0, 0]$. Whence $E = H^0(E)[0]$. We conclude $H^0(E) = f_* \mathcal{F}$ is finite locally free by More on Algebra, Lemma 0658 (and the characterization of finite projective modules in Algebra, Lemma 00NX). Commutation with base change means that $g^* f_* \mathcal{F} = f'_* (g')^* \mathcal{F}$ for a diagram as above and it follows from the already established commutation of base change for E. $\square$